Detection of Deepfakes in Images Using Convolutional Neural Networks

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Abstract. This study proposes a deepfake image detection model that combines MobileNetV2 with a frequency-domain analysis module to improve performance. The fusion of spatial and frequency features, along with an attention mechanism, enables the model to detect subtle manipulation artifacts. Experimental results demonstrate high accuracy and robustness, confirming the model's effectiveness and efficiency for deployment on resource-limited platforms.

Keywords: Fake Detection · Authenticity verification · MobileNetV2 · Classification · Deep learning.

In the digital age, multimedia content has expanded rapidly. However, sophisticated techniques such as deepfakes have emerged, using artificial intelligence, particularly deep neural networks, to create fake content nearly indistinguishable from the real. This poses ethical, legal, and security risks, as deepfakes can spread misinformation, influence public opinion, or enable fraud. It is therefore essential to develop effective methods to detect and mitigate the impact of such artificially generated content, i.e., determining whether an image has been created or altered using deepfake techniques [1].

As deep-fakes become increasingly advanced and harder to detect, robust and accurate methods are needed to distinguish authentic from manipulated content. This is especially critical in areas like news verification, national security, and privacy protection. Reliable detection is the key to preserving the integrity of information and preventing potential social and economic harm.

A major challenge in detecting deepfakes is the rapid evolution of generation techniques. Traditional methods relying on visual inconsistencies quickly become outdated as deep-fake creators refine their tools. This has driven the need for more advanced solutions, such as Deep Learning models capable of identifying complex, subtle patterns that are difficult to spot manually.

Deep Learning, especially Convolutional Neural Networks (CNNs), has shown strong performance in image classification and anomaly detection [6].. However, its use in deepfake detection poses challenges: models must generalize across diverse deepfake types and operate efficiently in real-time. Limited labeled datasets also hinder performance.

This work proposes a Deep Learning-based method to address these issues. By combining neural network architectures with data augmentation, we enhance generalization and efficiency. Our approach focuses on detecting artifacts and subtle signs typical of deepfakes, such as irregular lighting, textures, or facial details. The goal is to build a robust system that adapts to evolving deepfake techniques and supports efforts against digital misinformation.

1 State of the Art

Deepfake detection has advanced alongside increasingly realistic synthesis techniques. Dang et al. [1] introduced the Diverse Fake Face Dataset (DFFD), consisting of 2.6 million images (0.8M real and 1.8M fake), and designed an attention-guided CNN that optimizes both binary classification and an attention-map loss to localize manipulations. Their dual-loss model achieves top accuracy and precise tampering localization, but it incurs high computational costs and depends on dataset diversity for generalization.

Nguyen et al. [2] survey image- and video-based methods, ranging from CNNs that detect GAN artifacts (e.g., inconsistent eye details or lighting) to LSTMs that capture temporal anomalies. These approaches scale to large datasets, and some can adapt to new deepfake generators without retraining. However, they require extensive annotations and remain vulnerable to adversarial attacks.

Hsu and Kuo's Common Fake Feature Network (CFFN) uses a Siamese DenseNet with contrastive loss to discover shared manipulation signatures in a two-stage process—feature mining followed by classification—which generalizes to unseen GANs and provides visual explanations [3]. However, it relies on large training sets and is limited to low resolutions (64×64, 128×128).

Huang et al. [5] propose FakeLocator, combining spatial attention and feature enhancement to pinpoint altered regions in video frames, demonstrating robustness under compression at the expense of increased inference time. Afchar et al. [6] introduce MesoNet, a lightweight CNN focusing on mesoscopic features for low-resolution content, suitable for edge deployment but less sensitive to subtle or high-resolution forgeries.

These works highlight trade-offs among accuracy, localization, efficiency, and generalization, and point to ongoing challenges in scalability, robustness, and explainability.

2 Dataset Characteristics

Source: Deepfake and real images

The "deepfake and real images" dataset brings together human face photographs—both genuine and algorithmically manipulated deepfakes—with the explicit purpose of training, validating, and evaluating facial forgery detection systems under realistic conditions. Each image is stored as a 256 × 256-pixel JPEG, ensuring that models receive uniformly sized inputs suitable for deep learning architectures.

Additional information, such as demographic data (age, gender, ethnicity), details of fake generation (type of GAN, postprocessing, data origin) and ethical or legal aspects (consent, license, data protection) can be found on the official data set page at the link above.

Within this dataset, there are exactly two labels—"Real" and "Fake"—making the task a straightforward binary classification. The data are split into three disjoint subsets: 140,002 images (a balanced mix of real and fake) for training; 39,428 images for validation, used to tune hyperparameters and prevent overfitting; and a held-out test set of 10,905 images for final performance assessment. Each subset maintains roughly equal proportions of real and manipulated faces. The files are neatly organized into directories 'train', 'validation', and 'test', each containing the 'real' and 'fake' subfolders. This structure makes it easy to integrate the data into standard deep learning pipelines and to compare different detection models on a level playing field.

(a) Fake (b) Real

Fig. 1: Examples of a fake and real images from the dataset.

3 Methodology and Model Architecture

3.1 Contribution of the Proposed Solution in Relation to the State of the Art

The proposed model integrates innovative elements from current state-of-the-art methodologies, addressing key challenges in deepfake detection through a hybrid approach that combines spatial and frequency-based analyses.

Firstly, drawing inspiration from works such as the Diverse Fake Face Dataset (DFFD) [1] and FakeLocator [5], the model incorporates a specialized attention module that enhances the network's ability to highlight informative regions within facial images. Unlike traditional CNN architectures, the attention mechanism enables the network to dynamically prioritize areas potentially indicative of manipulation. This not only improves binary classification accuracy but also increases the interpretability of model predictions by localizing manipulated regions.

Secondly, similar to methods that utilize frequency-domain features to detect inconsistencies introduced by generative adversarial networks (GANs) [3] [4] ,

the proposed model employs a FrequencyAnalyzer component to capture subtle manipulation artifacts that are often imperceptible in the spatial domain. By applying a 2D Fourier transform and analyzing frequency magnitude maps, the model effectively identifies irregularities characteristic of digitally manipulated faces. This sets it apart from purely spatial-domain methods such as MesoNet [6], offering improved sensitivity to complex forgeries.

Additionally, the model leverages the robustness and computational efficiency of MobileNetV2 as its backbone [1] [4] [6], addressing common limitations found in more resource-intensive architectures. A two-phase training strategy—initially training only the classification head followed by comprehensive fine-tuning—optimizes both accuracy and computational resource usage. This approach reduces reliance on large training datasets and extensive computing power, as noted in previous works [3][4][6].

The model's dual representation, combining spatial attention and frequency-domain analysis, provides a balanced approach that delivers high detection accuracy while maintaining robustness against various manipulation types. Furthermore, the implementation of an adaptive threshold optimization mechanism based on the F1-score further enhances performance by enabling the model to dynamically adapt to data variability, thus improving generalization.

Collectively, these innovations address key limitations identified in the literature, positioning the proposed model as a robust, interpretable, and resource-efficient solution for detecting digitally manipulated facial content.

3.2 Block Diagram of the Model

The proposed architecture for deepfake detection combines visual and frequency features in a dual-branch structure. As illustrated in Figure 2, the input image undergoes initial preprocessing before being passed to two parallel streams: a CNN-based visual feature extractor and a frequency-domain analysis module. Features extracted from both branches are processed through an attention mechanism that assigns adaptive weights based on the extracted signals. These weighted features are then concatenated into a unified representation, which is subsequently passed through fully connected layers to classify the image as either real or fake.

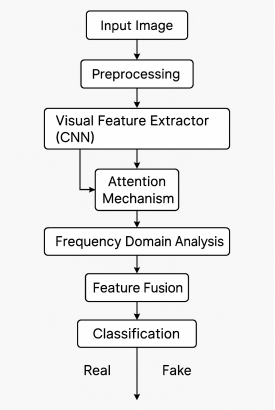


Fig. 2: Block diagram of the proposed architecture for deepfake detection combining visual and frequency features.

3.3 Model Process

The model follows the sequential steps outlined below:

1. **Data Preparation**: The dataset is divided into training, validation, and test sets, each containing clearly labeled "real" or "fake" images to support effective training and reliable evaluation.
2. **Image Preprocessing**: All images are resized to a fixed resolution and normalized to a standard scale, ensuring input consistency and training stability.
3. **Frequency Domain Analysis**: The FrequencyAnalyzer converts grayscale images to the frequency domain using a 2D Fast Fourier Transform (FFT), allowing the model to detect manipulation artifacts not visible in the spatial domain.
4. **Feature Extraction (MobileNetV2)**: A pretrained MobileNetV2 extracts spatial features from input images. During the initial training phase, most layers remain frozen to preserve learned representations from ImageNet.
5. **Attention Mechanism**: An attention module applies global average pooling to compute attention weights that modulate the contribution of feature regions before fusion.
6. **Feature Fusion**: Features from the spatial and frequency branches are concatenated to form a richer representation, improving the model's ability to distinguish between real and manipulated content.
7. **Classification Head**: The fused features are passed through dense layers with sigmoid activation to produce the final binary classification output.
8. **Two-phase Training Strategy**:
   - **Phase 1 (Initial Training)**: Only the classification head is trained, while the backbone remains frozen to focus on learning the decision boundary.

- **Phase 2 (Fine-tuning)**: All model layers are unfrozen and jointly trained to refine both spatial and frequency-based representations.

9. **Threshold Optimization**: After training, the decision threshold is optimized based on the F1-score to improve classification performance and adapt to data variability.
10. **Model Evaluation**: Final evaluation is performed using accuracy, the confusion matrix, and classification metrics to assess the model's effectiveness in detecting manipulated facial images.

3.4 Computational Performance Metrics

To evaluate the model's efficiency and potential for deployment on resource-constrained platforms, we analyzed its computational performance.

The final architecture comprises 2,422,881 parameters (9.24 MB), of which approximately 2 million are trainable. The model was trained on Google Colab using a shared GPU. Training was divided into two phases: during the fine-tuning phase, each epoch took between 442 and 841 seconds. Average inference time per image was approximately 25–41 milliseconds, depending on the batch size and GPU availability.

These results suggest that the Structured Model, despite its dual-branch architecture and integrated attention mechanism, remains computationally lightweight. Its relatively small size and fast inference make it suitable for deployment in mobile or embedded systems.

Table 1: Model Performance Metrics

Metric	**Value**
Total Parameters	2,422,881 (9.24 MB)
Trainable Parameters	2,027,457
Non-trainable Parameters	395,424
Average Inference Time	25–41 ms/image
Training Time per Epoch	442–841 s
Validation Accuracy (Final)	97.3%
Total Processing Time	Approximately 30 minutes

3.5 Comparative Analysis: MobileNetV2 Model vs. Structured Model

The following table presents a detailed architectural and methodological comparison between the Basic Model and the Structured Model implemented for deepfake detection. It highlights key differences across several critical components, including data loading strategies, preprocessing procedures, backbone architectures, additional frequency and attention modules, training protocols, optimization techniques, and evaluation metrics.

Table 2: Architectural Comparison: Basic Model vs. Structured Model

Aspect	**Basic Model**	**Structured Model**
Data Loading	Batch 16, size 224×224; Shuffle: Train (T), Val (F)	Batch 8, size 224×224; Shuffle: Train only
Preprocessing	Implicit norm. via MobileNetV2 pipeline	Explicit rescaling (1./255) before feature extraction
Backbone	MobileNetV2 (no top); GAP + Dropout(0.2) + Dense(1, sigmoid)	MobileNetV2 + Dense(128, ReLU) after GAP; optional concat. with freq. branch
Layer Freezing	All layers frozen	First 100 layers frozen; then full unfreeze
Additional Modules	None	**FreqAnalyzer**: FFT → log-mag. → Conv2D(16) + BN; **Attention-Module**: channel-wise via dense layers
Optimizer & LR	Adam (1e-4)	Phase 1: Adam (1e-3); Phase 2: Adam (1e-5)
Training Phases	1 phase (5 epochs)	Phase 1 (3 epochs); Phase 2 (2 epochs)
Callbacks	EarlyStopping (val_loss, patience 2)	EarlyStopping (pat. 4, restore weights); ReduceLROnPlateau (0.5, pat. 2, min 1e-6)
Monitored Metrics	Acc., Prec., Rec.	Accuracy only
Final Evaluation	Classif. report + conf. matrix (val set)	Classif. report + conf. matrix (test set); threshold search (F1); prob. histogram
Visualization Combined acc.	Acc. loss (2 phases); heatmap; prob. histogram	loss plots (5 epochs); heatmap

This comparative analysis enables a clear understanding of how the Structured Model builds upon and improves the Basic Model by integrating advanced modules such as frequency-domain analysis and attention mechanisms. These enhancements are designed to potentially improve the overall detection performance and model robustness.

4 Results

4.1 Results of the Structured Model

This section presents and analyzes the performance of the proposed Structured Model. We examine the evolution of training and validation metrics—specifically, accuracy and loss—across training epochs. This analysis is crucial for evaluating the effectiveness of the architecture, which incorporates spectral analysis and channel-wise attention mechanisms, as supported by prior studies highlighting their value in enhancing detection accuracy [1,5,2].

Additionally, we assess the model's generalization on unseen data to evaluate its robustness. This is a critical requirement emphasized in the literature for addressing the continuous evolution of deepfake generation techniques [4].
The following figure illustrates the training and validation performance of the Structured Model, showing accuracy (left) and loss (right) across epochs. The training accuracy (blue line) steadily increases, approaching 99%, indicating strong feature learning. Validation accuracy (red line) also improves consistently,

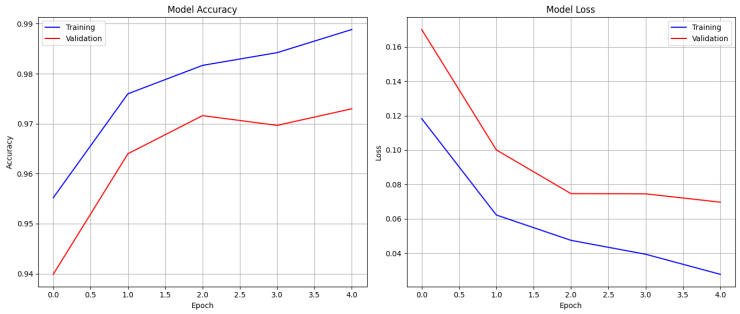


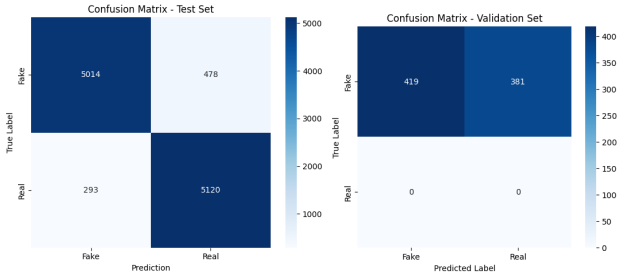
Fig. 3: Training and validation accuracy (left) and loss (right) curves of the Structured Model.

surpassing 97%, suggesting good generalization and reliable performance on unseen data [4].

The right panel shows the evolution of the loss metric, indicating a consistent decrease in loss values for both the training and validation datasets. Initially, the loss starts at relatively high values but rapidly decreases and stabilizes toward the end of the training process. Both training and validation losses converge to low values, suggesting that the model does not exhibit significant overfitting during the initial epochs. The small gap between the training and validation loss curves further supports the model's robustness and indicates strong generalization performance [2,9].

These preliminary results demonstrate that the proposed Structured Model—integrating spectral analysis and attention mechanisms—achieves both high accuracy and a substantial reduction in loss. This outcome validates the model's effectiveness in detecting manipulated deepfake images [1,5]. Moreover, the observed performance highlights the importance of combining spatial and spectral features, suggesting that the structured approach employed in this study significantly enhances the model's ability to detect subtle visual and frequency-based inconsistencies characteristic of deepfake imagery [3,2].

Confusion Matrix Analysis To gain deeper insight into the classification performance of the Structured Model, we present the confusion matrix derived from the test dataset. This analysis clearly identifies the counts of true positives, true negatives, false positives, and false negatives—key metrics for assessing the model's ability to distinguish between authentic and manipulated (deepfake) images [1,5,4].



(a) Confusion Matrix of the Structured Model on the Test Set.

(b) Confusion matrix of the Basic Model on the test set.

The confusion matrix presented in Fig. 4a shows the detailed classification results of the Structured Model. Specifically, out of 5,492 images labeled as *Fake*, the model correctly identified 5,014 (true negatives), achieving a detection rate of approximately 91.3%. It misclassified 478 fake images as *Real* (false positives), resulting in an error rate of about 8.7%. Similarly, among the 5,413 images labeled as *Real*, the model correctly identified 5,120 (true positives), yielding a correct classification rate of roughly 94.6%. Only 293 real images were incorrectly labeled as *Fake* (false negatives), corresponding to a misclassification rate of about 5.4%.

These results confirm that the Structured Model achieves high performance, accurately classifying the vast majority of images in both categories. Moreover, the low incidence of false positives and false negatives highlights the model's robustness and reliability in distinguishing real images from deepfakes [5,1]. This strong discriminative capability reinforces the model's potential for real-world deepfake detection, effectively reducing the risk of misclassification [4].

4.2 Distribution of Predicted Probabilities

To further assess the performance of the Structured Model, we analyzed the distribution of predicted probabilities on the test set. This analysis provides insight into the model's confidence when distinguishing between real and fake images [2,9].

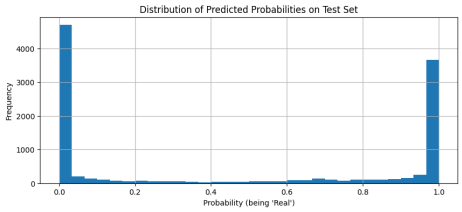


Fig. 4: Distribution of predicted probabilities for the "Real" class on the test set

Figure 4 shows a histogram of the probabilities assigned by the Structured Model to the "Real" class across all test samples. The x-axis represents the predicted probability of being "Real," while the y-axis indicates the number of images within each probability range.

The distribution is heavily concentrated at both extremes (near 0 and 1), indicating that the model made most predictions with high confidence. Specifically, a large number of fake images received probabilities close to 0, while real images were assigned probabilities close to 1. Few predictions fall within the intermediate range (e.g., 0.4–0.6), suggesting that the model is highly decisive and has learned a strong separation boundary between the two classes [2,9]. This confident prediction pattern further supports the model's robustness and reliability in real-world deepfake detection tasks [4].

Table 3: Deepfake Detection Model Performance at Different Thresholds

Threshold	Class	Precision	Recall	F1-score	Support
0.50	Fake	0.91	0.94	0.92	5492
	Real	0.94	0.90	0.92	5413
0.22	Fake	0.94	0.91	0.93	5492
	Real	0.91	0.95	0.93	5413

When using the default threshold of 0.50, the model achieves a balanced F1-score of 0.92 for both Fake and Real classes, with slightly higher recall for Fake (0.94) and higher precision for Real (0.94). By lowering the decision threshold to the optimal 0.22 (selected to maximize overall F1), both classes see an F1 improvement to 0.93. This adjustment trades a small drop in Fake recall (from 0.94 to 0.91) for a gain in Fake precision (from 0.91 to 0.94), and similarly shifts Real metrics (precision from 0.94 down to 0.91, recall from 0.90 up to 0.95), resulting in better overall balance.

4.3 Comparative Evaluation: Basic Model vs. Structured Model

To contextualize the improvements introduced by the Structured Model, it is essential to compare its performance with a baseline architecture, referred to as the Basic Model. This simpler model uses a standard convolutional backbone without frequency analysis or attention mechanisms and serves as a reference point for evaluating the impact of the proposed modifications [6,2,4].

Figure 5 presents the accuracy and the loss curves for the Basic Model.

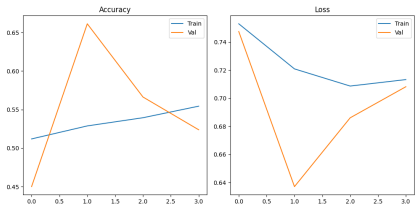


Fig. 5: Training and validation accuracy and loss curves for the Basic Model.

As shown in the left panel, training accuracy begins at approximately 51.7% and increases slowly, reaching only 55.6% by the final epoch. Validation accuracy peaks at 66.3% during the second epoch but drops to 52.9% by the end of training, indicating that the model struggles to maintain generalization and likely suffers from instability or underfitting [6]. In contrast, the Structured Model maintains a significantly higher and more consistent validation accuracy—above 97% throughout training—as previously shown in Figure 3 [8,4].

In the right panel of Figure 5, the loss curves show that validation loss decreases after the first epoch but rises again, ending above 0.70. This divergence from the training loss, which continues to decline slightly, suggests potential overfitting [5], despite the model's generally low accuracy. By comparison, the Structured Model exhibits smoothly decreasing and converging loss curves, reflecting a more stable and efficient learning process [1,8].

To further examine the classification behavior of the Basic Model, its confusion matrix is presented in Figure 4b.

The matrix shows that the model correctly classifies 419 fake images but misclassifies 381 fake samples as real. More critically, it entirely fails to recognize real images: both true positives and false negatives are zero. This reveals a severe prediction bias, where the model consistently defaults to the "Fake" class regardless of input [6,9]. In contrast, the Structured Model, as shown earlier in Figure 4a, correctly identifies 5,014 fake images and 5,120 real images, achieving a far more balanced and accurate classification [4,2].

While the Basic Model exhibits low overall accuracy and limited generalization, the Structured Model achieves significantly higher confidence in its predictions. As shown in Figure 4, most of its outputs are concentrated near probability values of 0 or 1, indicating decisive classification. Although the Basic Model's

confidence distribution is not explicitly visualized, its poor performance suggests it likely generates many uncertain or miscalibrated predictions, contributing to its instability [2,5]. .

5 Conclusion

This work demonstrates that integrating MobileNetV2 with frequency-domain analysis and attention mechanisms provides a high-performance and interpretable solution for deepfake detection. The model balances accuracy and computational efficiency, contributing to the development of robust forensic tools for real-world applications.

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References

1. Dang, H., Liu, F., Stehouwer, J., Liu, X. & Jain, A. On the detection of digital face manipulation. *Proceedings Of The IEEE/CVF Conference On Computer Vision And Pattern Recognition*. pp. 5781-5790 (2020
2. Nguyen, T., Nguyen, Q., Nguyen, D., Nguyen, D., Huynh-The, T., Nahavandi, S., Nguyen, T., Pham, Q. & Nguyen, C. Deep learning for deepfakes creation and detection: A survey. *Computer Vision And Image Understanding*. **223** pp. 103525 (2022)
3. Hsu, C., Zhuang, Y. & Lee, C. Deep fake image detection based on pairwise learning. *Applied Sciences*. **10**, 370 (2020)
4. Sultan, D. & Ibrahim, L. A comprehensive survey on deepfake detection techniques. *International Journal Of Intelligent Systems And Applications In Engineering*. **10**, 189-202 (2022)
5. Huang, Y., Juefei-Xu, F., Guo, Q., Liu, Y. & Pu, G. Fakelocator: Robust localization of gan-based face manipulations. *IEEE Transactions On Information Forensics And Security*. **17** pp. 2657-2672 (2022)
6. Afchar, D., Nozick, V., Yamagishi, J. & Echizen, I. Mesonet: a compact facial video forgery detection network. *2018 IEEE International Workshop On Information Forensics And Security (WIFS)*. pp. 1-7 (2018)
7. Ahmed, S., Sonuç, E., Ahmed, M. & Duru, A. Analysis survey on deepfake detection and recognition with convolutional neural networks. *2022 International Congress On Human-Computer Interaction, Optimization And Robotic Applications (HORA)*. pp. 1-7 (2022)
8. Soudy, A., Sayed, O., Tag-Elser, H., Ragab, R., Mohsen, S., Mostafa, T., Abohany, A. & Slim, S. Deepfake detection using convolutional vision transformers and convolutional neural networks. *Neural Computing And Applications*. **36**, 19759-19775 (2024)
9. Almars, A. Deepfakes detection techniques using deep learning: a survey. *Journal Of Computer And Communications*. **9**, 20-35 (2021)